Estimator forensics for the wide-binary gravity test: the eccentricity–triple coupling and a corrected Gaia DR3 application

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Data-provenance correction to versions 1–2

An independent reader alerted the author that the 81,088-pair input used for the observational analysis in versions 1–2 was named `Newton_dr3_MSMS_d200pc_5.csv`. A subsequent audit confirmed that the redistributed reduction matches Chae's official virtual Newtonian v5 catalogue, rather than the real Gaia DR3 catalogue. Consequently, all observational results in the original Section 5 are withdrawn, including the quoted $\gamma = 1.05$, $\Delta \ln \mathcal{L} = 129$, the associated $\sim 16\sigma$ statement, and the claimed deep-bin exclusion. They cannot be interpreted as measurements of the sky. This revision uses the complete 81,880-pair real catalogue with corrected RUWE from Zenodo record 10986733, separates the surviving synthetic result from the corrected observational application, and reports failed validation controls rather than overriding them. Full forensic details are supplied with the reproducibility package.

Abstract

We study the sensitivity of wide-binary gravity estimators to residual hierarchical triples and eccentricity modelling. Known-truth catalogues are constructed from Keplerian binaries and hierarchical triples, with Gaia-like noise and the upstream velocity selection. On Newtonian truth with a residual triple fraction of 0.2–0.3, a full-median estimator recovers $\gamma = G_{\rm eff}/G_{\rm N} = 1.062$–$1.128$, whereas a tail-truncated estimator gives 1.002–1.044 and a binary–triple mixture likelihood gives 0.98–1.00. When $\gamma = 1.4$ is injected above 2 kau, the latter two estimators give 1.306–1.346 and 1.36–1.40. This supports a methodological claim: residual triple tails can create a positive bias in a full-median estimator, while matched truncation or explicit mixture modelling reduces it. The corrected real-data application uses Chae's 81,880-pair Gaia catalogue with corrected RUWE. In the 2–30 kau zone, the truncated estimator gives 1.084 $[1.063, 1.105]$ for empirical per-zone eccentricities but 1.007 $[0.987, 1.023]$ for a thermal distribution; the corresponding mixture fits give 1.13 and 1.01. Crucially, the empirical and superthermal branches fail the pre-specified 2 per cent validation-zone control, while the thermal branch passes. Deep-bin estimates span 1.22–1.31 with broad 68 per cent intervals. The corrected Gaia DR3 outcome is therefore model-dependent and non-conclusive under the pre-registered decision logic. No high-significance observational gravity verdict is claimed.

1 Introduction

Wide binaries probe internal accelerations near and below $a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. Some MOND external-field calculations predict $\gamma \simeq 1.35$–1.40, corresponding to an approximately 18 per cent velocity increase, whereas Newtonian gravity predicts $\gamma = 1$. Analyses of Gaia DR3 have reached conflicting conclusions (e.g. Chae 2023, 2024; Banik et al. 2024; Pittordis et al. 2025). Selection, unresolved multiplicity, eccentricity distributions, and perspective corrections can all be coupled to the inferred gravity parameter.

This paper has two distinct aims. First, we perform an injection–recovery comparison of estimator families on synthetic universes of known truth. Second, we apply the same pipeline to a corrected real Gaia catalogue while enforcing a validation-zone control. These aims must remain separate: success on a simulated Newtonian catalogue is a calibration result, not a measurement of the gravitational field in nature.

2 Catalogue provenance and audit

The original scripts read an 81,088-row file derived from `Newton_dr3_MSMS_d200pc_5.csv`. Chae's public Zenodo records identify the `Newton_*` family as virtual catalogues whose astrometric distances and proper motions are replaced by simulated Newtonian orbits and hidden companions. Source identifiers and several static columns remain tied to the real catalogue, which made a superficial identifier match insufficient to establish observational provenance.

Our audit joins the repository reduction to both the official v5 mock and the real catalogue by the two Gaia source identifiers. All 81,088 rows and all 19 analysis columns of the reduction agree with the v5 mock at relative tolerance 10^{-9}. As an independent fingerprint, the median $|d_1 - d_M|$ is 0.000768 pc in the legacy file and 0.137569 pc in the real catalogue; 94.19 per cent of legacy pairs, but only 8.42 per cent of real pairs, lie within the projected orbital line-of-sight offset. The relative proper motions have correlation 0.626 with their real-catalogue counterparts.

For the corrected analysis we use the full `gaia_dr3_MSMS_d200pc_ruwe.csv` table from Zenodo record 10986733: 81,880 rows and MD5 `1b6c5063163a4e6c07043d13aeb70f55`. This release also corrects the previously mismatched RUWE information. The loader verifies the URL, filename, digest, schema and row count, and refuses both `Newton_*` inputs and the 81,088-row fingerprint.

3 Observable and corrected pipeline

The dimensionless projected velocity is

$$\tilde{v} = \frac{|\Delta v_\perp|}{\sqrt{G_\mathrm{N} M_\mathrm{tot}/s_\mathrm{2D}}}. \tag{1}$$

We use an angular perspective correction evaluated at the common mean distance. Missing radial velocities are explicitly masked, and radial-velocity uncertainty is propagated through the change of sky basis. A trial γ is applied to the model velocity before measurement noise, the upstream El-Badry-type velocity cut, and the estimator cut. In a hierarchical triple it scales the low-acceleration outer orbit only; compact inner photocentre motion remains Newtonian.

The fiducial selection requires distance below 200 pc, $R_\mathrm{chance} < 0.01$, RUWE below 1.4 for both components, and $\sigma_{\tilde{v}} < 0.10$. It retains 38,472 pairs. The validation zone $0.2 \le s_\mathrm{2D} < 2$ kau contains 28,221 pairs, the test zone $2 \le s_\mathrm{2D} < 30$ kau contains 7,900, and the deep bin $0.03 \le g_\mathrm{N}/a_0 < 0.3$ contains 403. A strict RUWE cut of 1.2 retains 31,264 pairs. The perspective correction is small in the validation zone but not for the widest systems: the 90th percentile of $|\Delta\tilde{v}|$ is 0.00172, 0.0286, and 0.224 in the validation, test, and deep zones respectively.

We compare three eccentricity families: empirical per-zone values supplied in the catalogue, thermal, and superthermal. The primary truncated estimator E2 uses medians below $T = \sqrt{2}$ in data and zone-matched noisy templates. The secondary estimator E3 maximizes a Poisson likelihood over $(\gamma, f_\mathrm{trip})$ for a binary plus hierarchical-triple mixture. Intervals reported for E2 are the central 68 per cent of 400 bootstrap resamples. E3 intervals use the profile-likelihood $\Delta \ln \mathcal{L} = 0.5$ contour. These are conditional intervals within each specified model, not a complete systematic uncertainty.

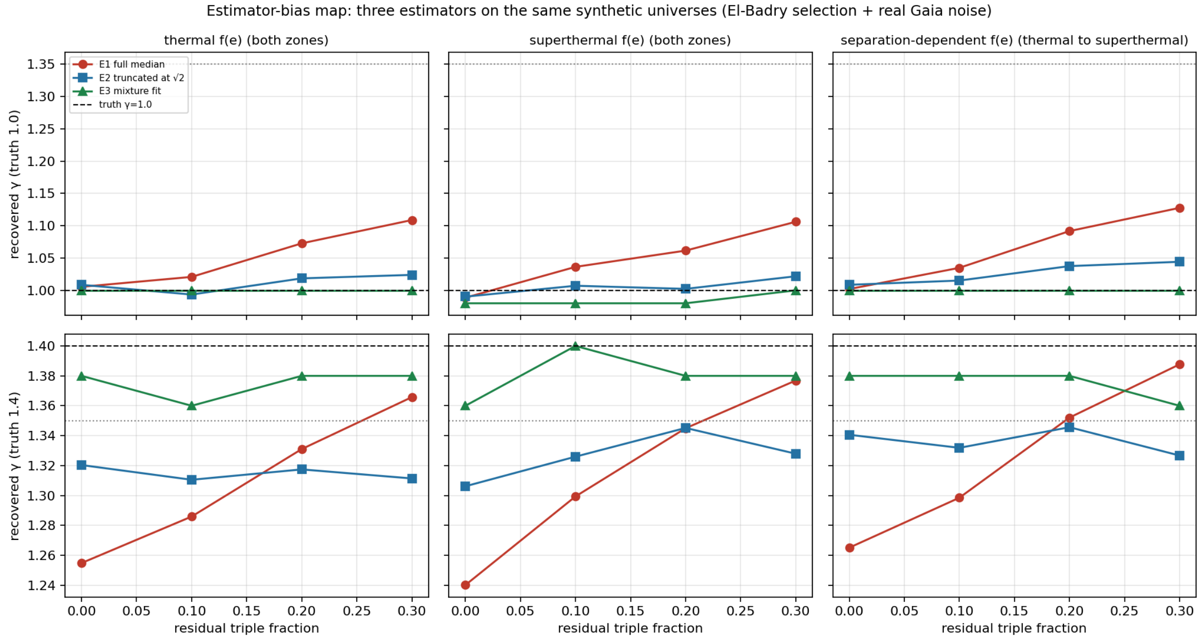

Figure 1: Corrected known-truth injection–recovery grid. The rows show Newtonian and $\gamma = 1.4$ truth; columns show the eccentricity prescriptions. The full-median estimator becomes positively biased as the triple fraction increases. The truncated and mixture estimators better separate the two gravity regimes.

4 Synthetic injection–recovery

The population engine follows the Keplerian binary and hierarchical-triple construction described in Pittordis et al. (2025), including photocentre motion and averaging over the Gaia DR3 time baseline. It reproduces the principal published validation statistics without tuning. Synthetic catalogues use noise resampled from the corrected real table. Common random orbital realisations are used across the gravity grid, and γ is applied before noise and selection.

We generate $\gamma \in \{1.0, 1.4\}$, three eccentricity prescriptions and $f_{\text{trip}} \in \{0, 0.1, 0.2, 0.3\}$. E1 is a validation-calibrated ratio of full medians; E2 applies the matched $\tilde{v} < \sqrt{2}$ truncation; E3 fits binary and triple histograms jointly.

For Newtonian truth at $f_{\text{trip}} = 0.2$–0.3, E1 gives 1.062–1.128, E2 gives 1.002–1.044, and E3 gives 0.98–1.00. For injected $\gamma = 1.4$, over all triple fractions, E2 gives 1.306–1.346 and E3 gives 1.36–1.40; E1 gives 1.240–1.388. Thus the synthetic experiment supports an estimator-bias diagnosis, but its single-seed grid and simplified triple population do not by themselves establish the astrophysical triple fraction or the gravity law.

5 Corrected Gaia DR3 application

The mandatory validation control compares the observed validation-zone median to its Newtonian template and requires agreement within 2 per cent. Table 1 shows that only the thermal branch passes. The empirical branch used for the original primary analysis misses the control by 10.99 per cent and must stop under the pre-registered logic.

The global E2 estimate changes little when the RUWE threshold is tightened: the empirical, thermal and superthermal values become 1.067, 0.996 and 0.996. Scanning $T = 1.2$–1.8 gives 1.084–1.098 for the empirical family and 1.001–1.021 for thermal/superthermal families. This threshold stability does not remove the much larger eccentricity-model dependence or repair a failed validation control.

Table 1: Corrected real-catalogue results at $T = \sqrt{2}$. Bracketed ranges are 68 per cent conditional intervals. A failed validation branch is reported for diagnosis, not used for a gravity verdict.

$f(e)$ model	validation offset	E2: 2–30 kau	E2: deep bin	E3: γ; f_{trip}
empirical	+10.99% (fail)	1.084 [1.063, 1.105]	1.308 [1.232, 1.418]	1.13 [1.13, 1.14]; 0.14
thermal	+1.77% (pass)	1.007 [0.987, 1.023]	1.221 [1.132, 1.316]	1.01 [1.00, 1.02]; 0.16
superthermal	+4.06% (fail)	1.007 [0.988, 1.022]	1.222 [1.146, 1.314]	1.04 [1.02, 1.05]; 0.15

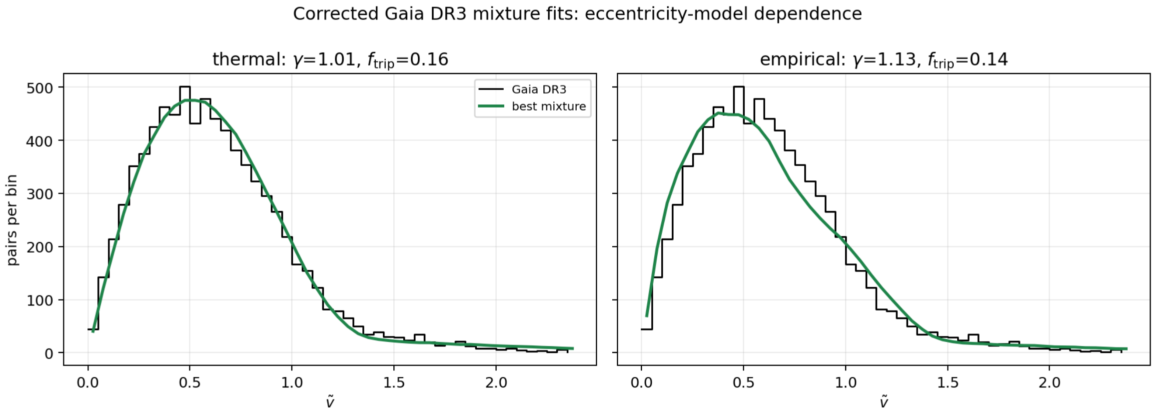

Figure 2: Corrected Gaia DR3 mixture fits for thermal (left) and empirical (right) eccentricity prescriptions. The difference between panels is a model sensitivity, not an observational detection.

The E3 best fits similarly depend on the eccentricity prescription. Although the conditional likelihood strongly disfavors $\gamma = 1.4$ in the thermal and superthermal implementations, quoting this as a model-independent significance would be inappropriate: the empirical branch shifts the best fit and fails the validation control, and the likelihood does not marginalize over the full eccentricity and triple-population uncertainty.

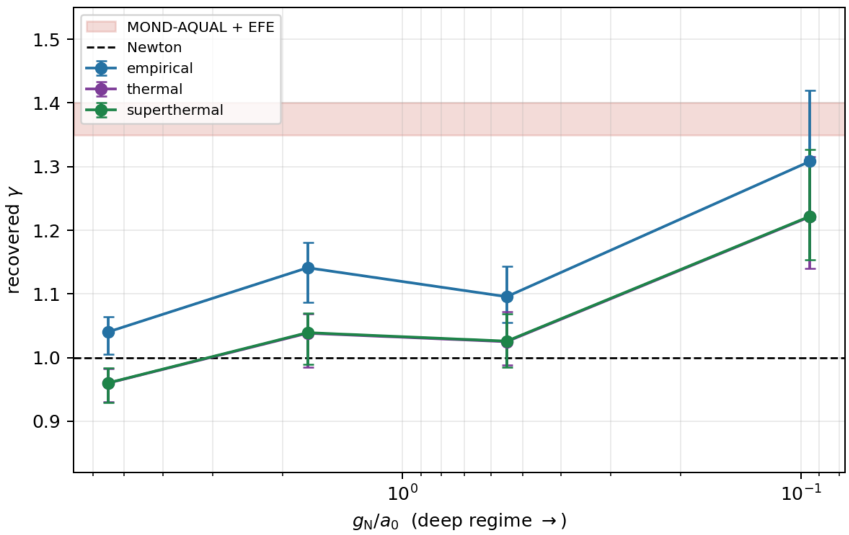

Figure 3: E2 estimates versus internal acceleration for the three eccentricity families. The divergence and broad deep-bin intervals motivate the non-conclusive classification.

In the deep bin, the 68 per cent ranges are 1.232–1.418 (empirical), 1.132–1.316 (thermal) and 1.146–1.314 (superthermal). Perspective effects are large there, and the sample contains only 403 pairs. These conditional results do not justify the previous claim that the MOND-EFE range is excluded. Taken together, the failed controls and model spread place the corrected DR3 application in the pre-declared grey/non-conclusive category.

6 Implications for the DR4 protocol

The original pre-registration file is retained unchanged as a historical record, but its DR3 calibration paragraph is invalid because it inherited the mock-catalogue analysis. A dated pre-DR4 amendment removes that calibration standard, adds mandatory source-identity and checksum checks, records the gravity-ordering correction for triples, and requires propagation of radial-velocity uncertainty. It does not use or anticipate a Gaia DR4 outcome.

The most important surviving rule is the STOP condition: a failed Newtonian validation zone is a pipeline result that must be published, not tuned away. The present exercise demonstrates why model families and validation outcomes must be shown individually rather than compressed into a nominal statistical significance.

7 Conclusions

Versions 1–2 confused a virtual Newtonian catalogue with observational Gaia data. Their Section 5 gravity measurement and significance claims are withdrawn. Reanalysis of the corrected real catalogue produces materially model-dependent results; only the thermal eccentricity branch passes the specified validation control, and the deep bin is not decisive. This work therefore makes no high-significance claim for either Newtonian or modified gravity from Gaia DR3.

The narrower synthetic conclusion survives. In known-truth experiments, residual triple tails bias a full-median estimator upward, while matched truncation and explicit mixture modelling reduce the bias and distinguish the simulated Newtonian and $\gamma = 1.4$ regimes. Whether those models adequately describe the sky remains a separate astrophysical question for better data and pre-registered validation with Gaia DR4.

Data and code availability

The corrected code, provenance audit, machine-readable outputs, legacy quarantine, and DR4 amendment are available at `https://github.com/HBoufourou/paperI-wide-binaries`. Versions 1–2 are archived at Zenodo DOI `https://doi.org/10.5281/zenodo.22114321`; they are affected by the error described here. The corrected release DOI will be added after a new Zenodo version is minted. The real Gaia-derived catalogue is downloaded from Zenodo record 10986733 and checksum-verified by the pipeline.

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Declaration on computational tools

Large language models were used as computational and analytical assistants to write, debug and audit scripts and text. The provenance finding was independently reproduced from public files and is recorded in executable code. The author is responsible for the assumptions, interpretation, correction and publication decisions.

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